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Packet Sniffing Using Raw Sockets

Aim:

To write a program using raw sockets to capture and analyse network packets and examine their basic header information.

Algorithm / procedure.

1. Import the required socket and packet processing libraries.
2. Create a raw socket using the appropriate address family and protocol.
3. Run the program with the necessary administrative privileges.
4. Bind the socket to the appropriate interface if required by the operating system.
5. Capture packets arriving at the selected interface.
6. Extract the Ethernet header when available followed by the IP header and relevant transport layer headers.

Sample output:

Packet Sniffer started...

Source IP : 192.168.1.10

Destination IP : 192.168.1.1

protocol : TCP

Packet length : 60 bytes

Source IP : 192.168.1.10

Destination IP : 8.8.8.8

protocol : UDP

Packet length : 72 bytes

Packet Capture completed.



7. Identify packet information such as Source IP address, destination IP address, protocol and packet length.

8. Display the extracted information in a readable format.

9. Continue capturing packets until the users stop the program.

10. close the raw socket and release the resources.

Result :-

A packet - sniffing program was implemented successfully using raw sockets. Network packets were captured and their basic header information including source address, destination address, and protocol was analysed.